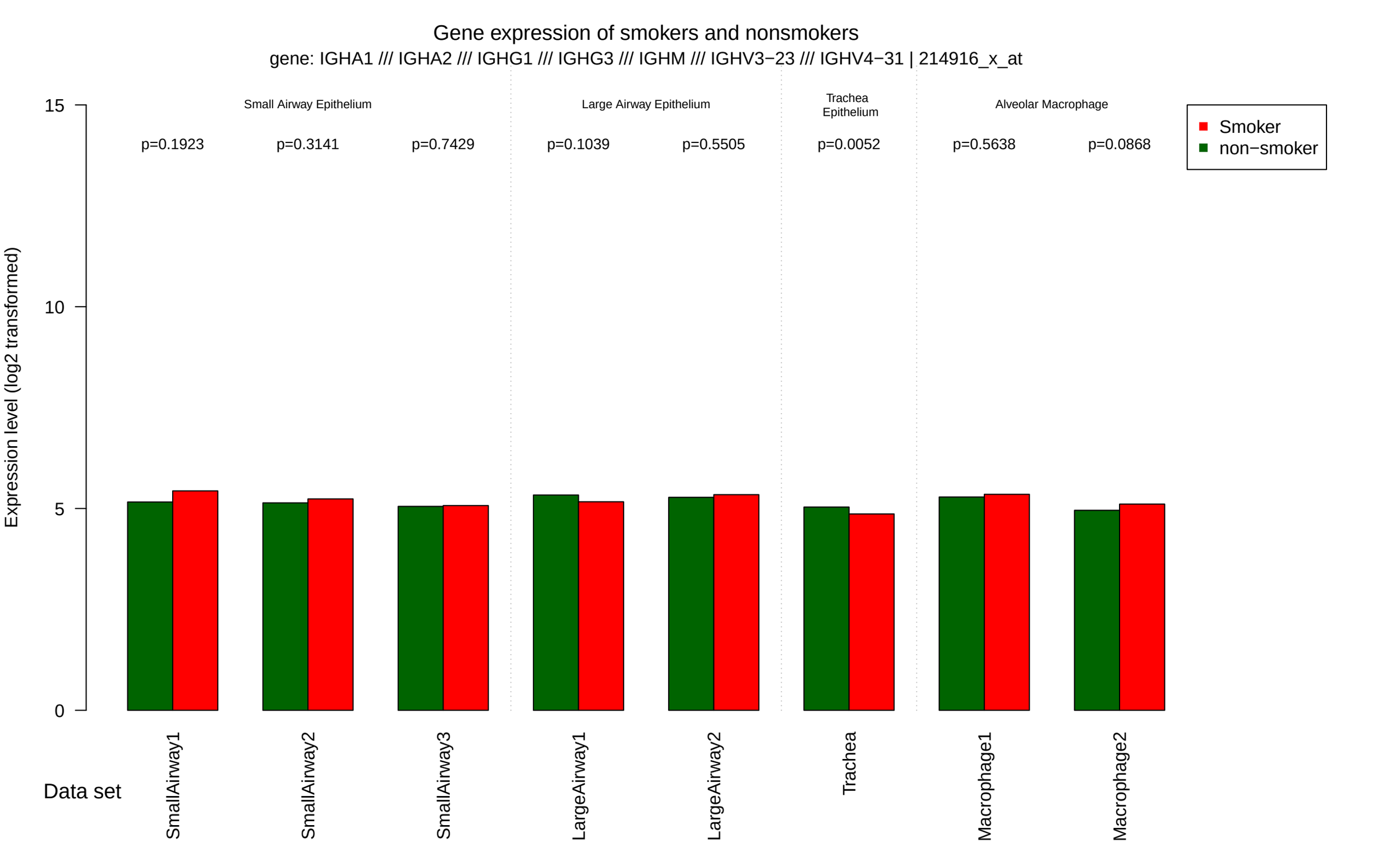
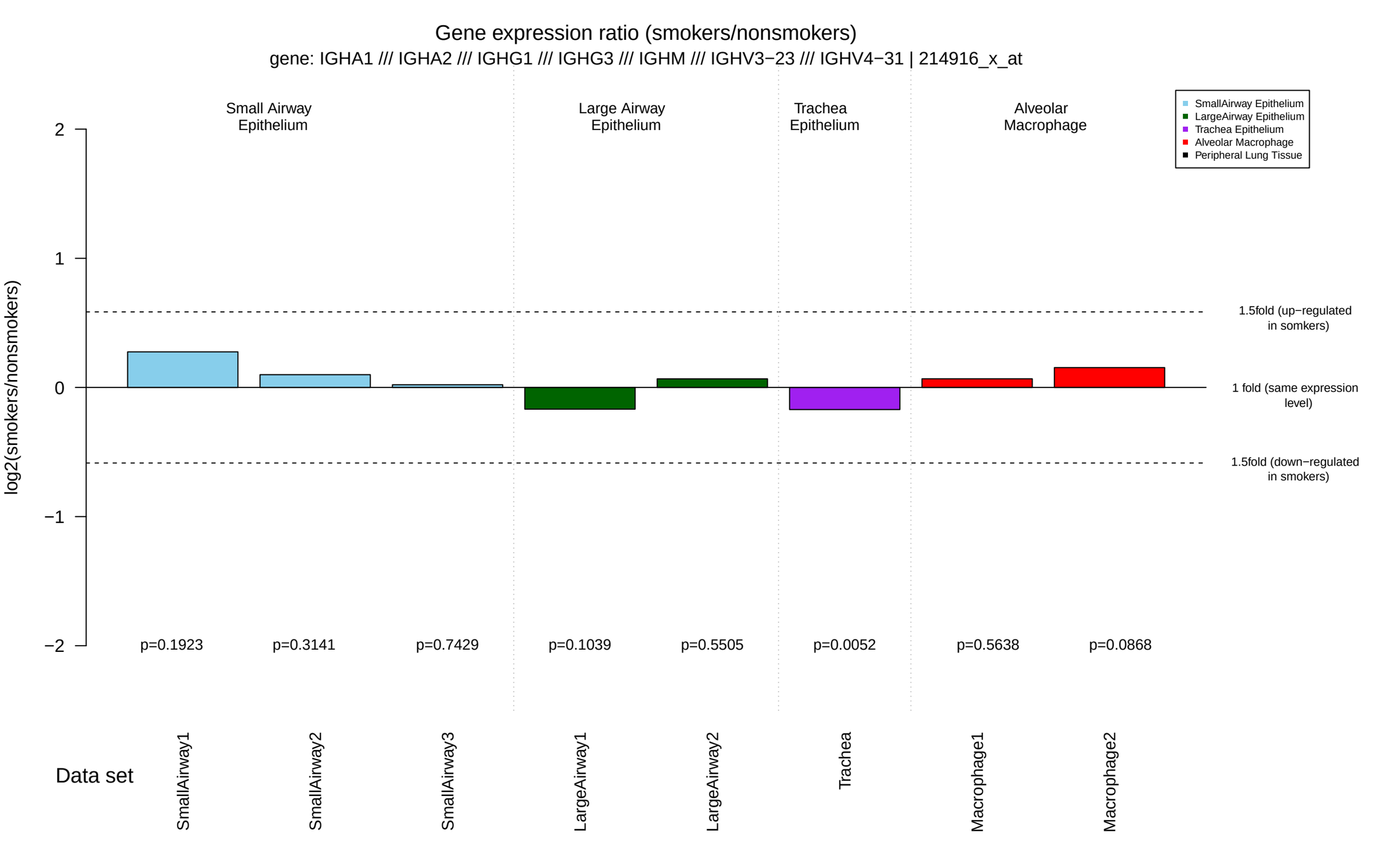
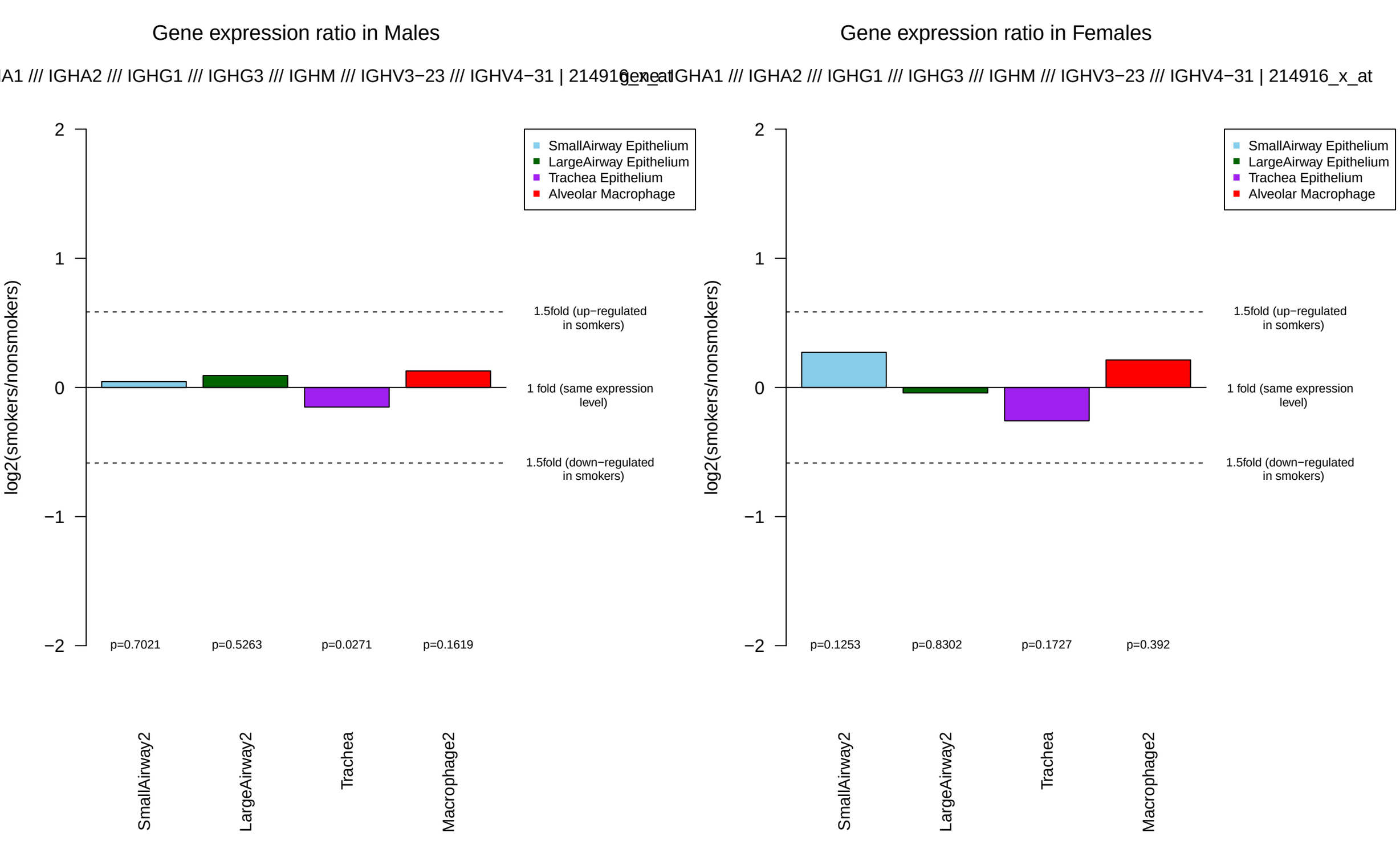
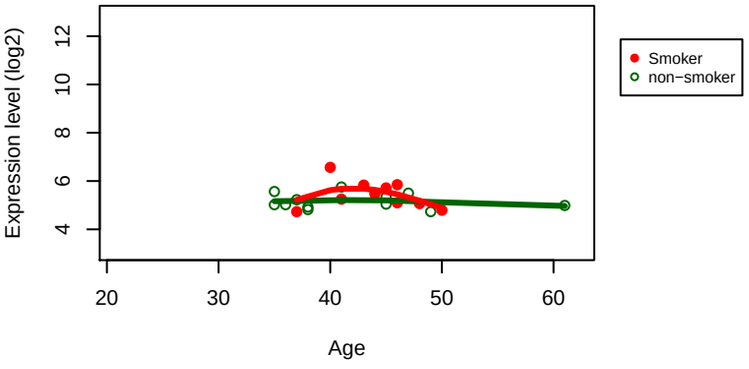

Gene expression of smokers and nonsmokers
gene: IGHA1 /// IGHA2 /// IGHG1 /// IGHG3 /// IGHM /// IGHV3−23 /// IGHV4−31 | 214916_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1923
p=0.3141
p=0.7429
p=0.1039
p=0.5505
p=0.0052
p=0.5638
p=0.0868
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

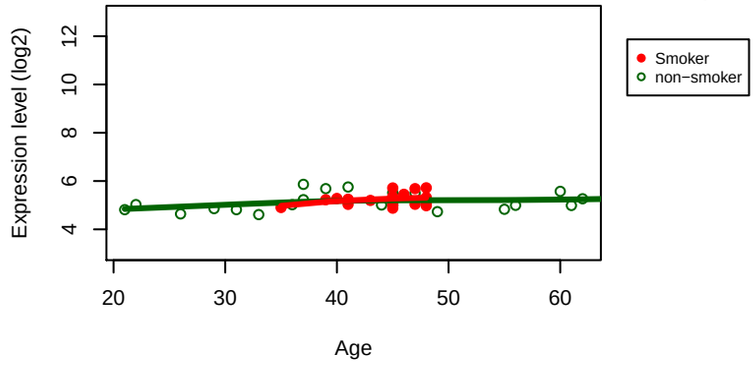

Gene expression ratio (smokers/nonsmokers)
gene: IGHA1 /// IGHA2 /// IGHG1 /// IGHG3 /// IGHM /// IGHV3−23 /// IGHV4−31 | 214916_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1923
p=0.3141
p=0.7429
p=0.1039
p=0.5505
p=0.0052
p=0.5638
p=0.0868
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2


Gene expression ratio in Males
A1 /// IGHA2 /// IGHG1 /// IGHG3 /// IGHM /// IGHV3−23 /// IGHV4−31 | 214916_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7021
p=0.5263
p=0.0271
p=0.1619
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
Gene_IGHA1 /// IGHA2 /// IGHG1 /// IGHG3 /// IGHM /// IGHV3−23 /// IGHV4−31 | 214916_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1253
p=0.8302
p=0.1727
p=0.392
SmallAirway2
LargeAirway2
Trachea
Macrophage2

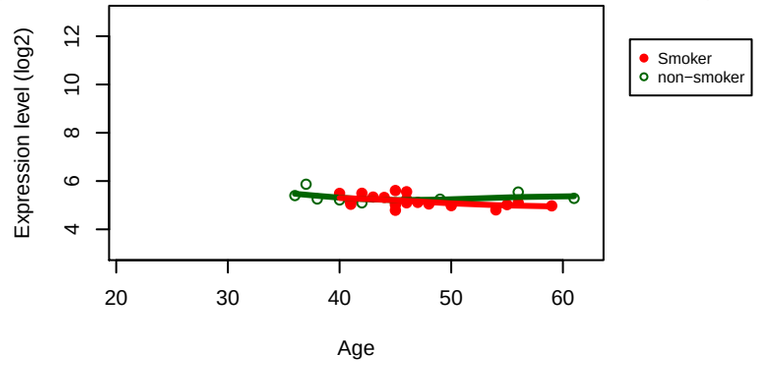
set: SmallAirway1 (p=0.1923)



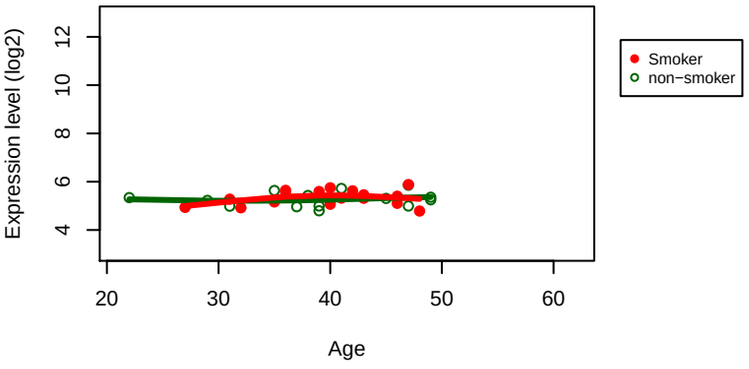
set: SmallAirway2 (p=0.3141)



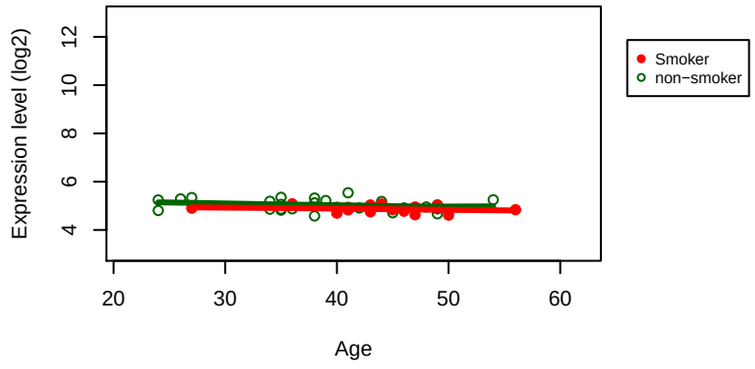
set: LargeAirway1 (p=0.1039)



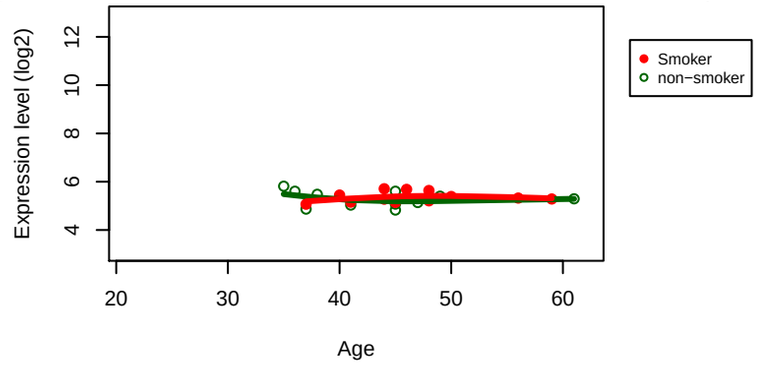
set: LargeAirway2 (p=0.5505)



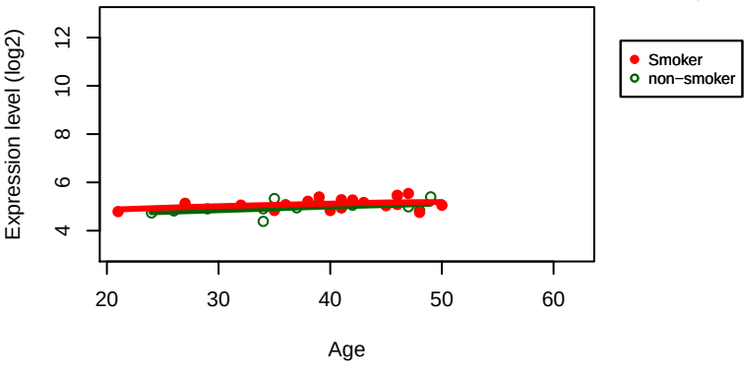
set: Trachea (p=0.0052)

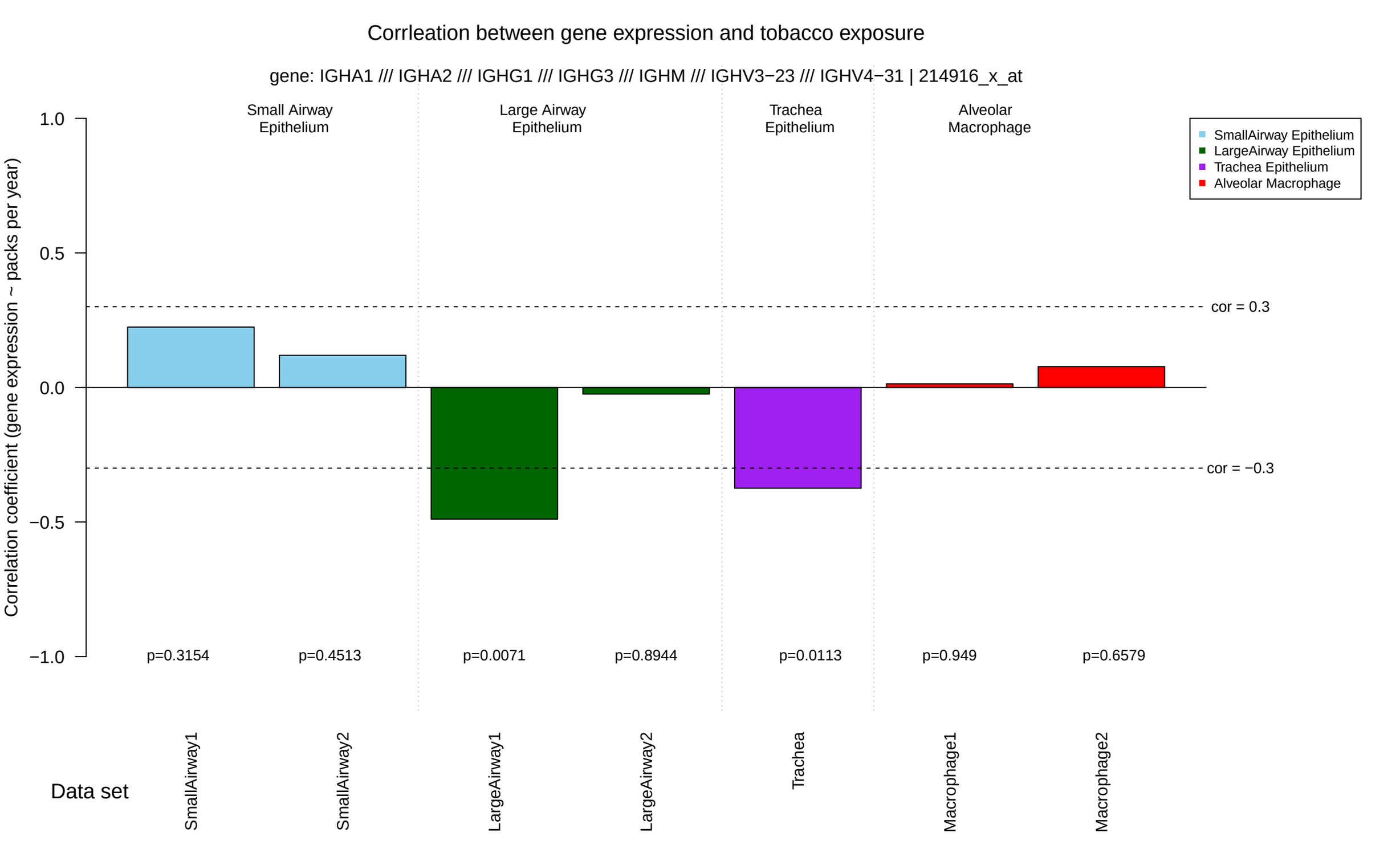


set: Macrophage1 (p=0.5638)



set: Macrophage2 (p=0.0868)




Corrleation between gene expression and tobacco exposure
gene: IGHA1 /// IGHA2 /// IGHG1 /// IGHG3 /// IGHM /// IGHV3−23 /// IGHV4−31 | 214916_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3154
p=0.4513
p=0.0071
p=0.8944
p=0.0113
p=0.949
p=0.6579
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2